

Aayush Kumar Singh

Full Stack Developer

aayushsinghrajput2002@gmail.com · +977 9805981168 · Kathmandu, Nepal · Portfolio · GitHub · LinkedIn

Professional Summary

Aspiring Full Stack Developer with internship experience building scalable web applications using React.js, Next.js, Node.js, Express.js, and FastAPI. Proficient in TypeScript, REST API design, and performance optimization. Experienced with AI-integrated systems using RAG pipelines, LLM integration, and vector databases. Hackathon runner-up.

Technical Skills

Languages: TypeScript, JavaScript, Python

Frontend: React.js, ReactNative, Expo, Next.js, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, FastAPI

Databases: MongoDB, PostgreSQL(Neon), Vector Databases (ChromaDB)

AI/ML: NLP, Retrieval-Augmented Generation (RAG), LLM Integration, scikit-learn, MobileNet, Semantic Search

Tools: Git, GitHub, Vercel, Render, Netlify, Docker, Inngest, Stripe, Jest, React Testing Library, Playwright

Concepts: REST API Design, JWT Authentication, Redux, Context API, SEO Optimization, Agile/Scrum, API Documentation

Experience

Web & Mobile Application Development Intern (AI Integration)

Jan 2026 – Apr 2026

Nexalaris Tech Pvt. Ltd.

- Led Team Sphinx (4 interns) delivering AI-integrated software in Agile/Scrum sprints with daily standups and stakeholder communication.
- Built *Himalayan Bodhi Spices* e-commerce platform with Next.js, TypeScript, Tailwind CSS, and Framer Motion, reducing layout inconsistencies across 10+ pages and improving mobile responsiveness.
- Designed REST API endpoints for core authentication and user management features (login, registration, profile), documented the complete REST API, and performed UI, functional, and regression testing on the Subhyatra React Native app, identifying and logging 20+ bugs across 3 release cycles.

Projects

Employee Management System

<https://ems-one-tawny.vercel.app/login>

React, Node.js, Express.js, MongoDB, Inngest, JWT, TypeScript, Tailwind CSS

- Full-stack HR platform covering attendance, leave workflows, and automated payslip generation for 50+ employee profiles; reduced manual HR effort by an estimated **60%**.
- Role-based JWT auth across 2 dashboards; 15+ RESTful API endpoints with clean frontend/backend separation; Inngest automation for alerts and email delivery via Nodemailer.

GreenCart- Full-Stack Grocery Web App

<https://green-cart-grocery-web-app.vercel.app>

React, Node.js, Express.js, MongoDB, Stripe, Cloudinary, Tailwind CSS

- Full-stack e-commerce grocery platform with end-to-end shopping workflows including cart management, multi-address support, and dual payment processing (Cash on Delivery + Stripe) for a seamless user experience.
- Role-based JWT cookie auth across 2 portals (User + Seller); 18+ RESTful API endpoints with clean frontend/backend separation; Cloudinary integration for dynamic product image uploads and Stripe webhooks for reliable payment verification.

SmartPrep AI – A Virtual Teaching Assistant

<https://github.com/AayushSinghRajput/SmartPrep-AI.git>

Next.js, FastAPI, MongoDB, ChromaDB, RAG, LLM, NLP, Cloudinary

- Built a full-stack learning platform with a Next.js frontend, FastAPI backend, and MongoDB — supporting PDF uploads, user sessions, multi-document ingestion, automated assessments, and progress tracking with full OpenAPI/Swagger documentation.
- Integrated RAG pipelines with ChromaDB vector search and LLM to convert uploaded PDFs into summaries, lessons, and MCQs, reducing manual study preparation time by **70%**.

TuitionMaster – A Tutoring Marketplace Platform

<https://www.tuitionmaster.guru>

React, Node.js, Express, MongoDB, Gemini AI, Cloudinary, Tailwind CSS, Docker

- Built a full-stack tutoring marketplace connecting students with verified tutors — featuring JWT + Google OAuth authentication, a 4-step profile wizard with Cloudinary-hosted avatar and CV uploads, guest reviews with admin moderation, and a separate admin panel for tutor verification and analytics.
- Integrated a Gemini-powered AI chat assistant using tool-calling to search tutors, fetch profiles, and post requirements from live data, enabling students to find matching tutors through natural-language queries and reducing tutor discovery time by **60%**.

Achievements

-
- **Runner-up – Janakpur Hackathon 2.0** (Dec 2025) — Built a civic-tech platform with an AI legal chatbot trained on Nepal Constitution 2072 using RAG, real-time notifications, and an NGO categorization system for incident response.

Education

Bachelor in Computer Engineering

2022 – 2026

Institute of Engineering, Purwanchal Campus